

Equations Requiring Four or More Steps

Name
Date
Course

$$\textcircled{1} \quad 4x + 5 - 13 - 2x + 5x + 7 = -6 + 9$$

$$7x - 1 = 3$$

$$1 + 7x - 1 = 3 + 1$$

$$7x = 4$$

$$\frac{7x}{7} = \frac{4}{7}$$

$$x = \frac{4}{7}$$

$$\textcircled{2} \quad 10 - 15 = 4y - 9y + 2 + 9y - 3 + 11$$

$$-5 = 4y + 10$$

$$-10 - 5 = 4y + 10 - 10$$

$$-15 = 4y$$

$$\frac{-15}{4} = \frac{4y}{4}$$

$$-\frac{15}{4} = y$$

$$\textcircled{3} \quad -\frac{2z}{7} - 3 + \frac{3z}{5} + 2 = -7 + 5 + 1$$

$$1 - \frac{2z}{7} + \frac{3z}{5} - 1 = -1 + 1$$

$$-\frac{2z}{7} + \frac{3z}{5} = 0$$

$$-\frac{10z}{35} + \frac{21z}{35} = 0$$

$$(35) \frac{11z}{35} = 0(35)$$

$$\frac{11z}{11} = \frac{0}{11}$$

$$z = 0$$

$$\textcircled{4} \quad -8 + 9 = -2x + 4 - 1 - 5x$$

$$1 = -7x + 3$$

$$-3 + 1 = -7x + 3 - 3$$

$$-2 = -7x$$

$$\frac{-2}{-7} = \frac{-7x}{-7}$$

$$\frac{2}{7} = x$$

$$\textcircled{5} \quad -10 + 7y - 12y + 3 - 2 + 4y = 7 - 8 - 5$$

$$-y - 9 = -6$$

$$9 - y - 9 = -6 + 9$$

$$-y = 3$$

$$\frac{-1y}{-1} = \frac{3}{-1}$$

$$\boxed{y = -3}$$

$$\textcircled{6} \quad -4 - 9 + 6 - 10 = -6 + \frac{3z}{4} - \frac{5z}{7}$$

$$-17 = -6 + \frac{21z}{28} - \frac{20z}{28}$$

$$6 - 17 = -6 + \frac{z}{28} + 6$$

$$-11 = \frac{z}{28}$$

$$(28)(-11) = \frac{z}{28}(28)$$

$$\boxed{-308 = z}$$

$$\textcircled{7} \quad 5 - 3x - 6x - 2 + 11x = -4 + 7 + (-8)$$

$$2x + 3 = -5$$

$$-3 + 2x + 3 = -5 - 3$$

$$2x = -8$$

$$\frac{2x}{2} = \frac{-8}{2}$$

$$\boxed{x = -4}$$

$$\textcircled{8} \quad -3 - 8 - 1 = -3 + 7 + 14y + 1 - 6 - 5y$$

$$-12 = 9y - 1$$

$$-12 = 9y - 1 + 1$$

$$-11 = 9y$$

$$\frac{-11}{9} = \frac{9y}{9}$$

$$\boxed{-\frac{11}{9} = y}$$

$$\textcircled{9} \quad -\frac{5z}{8} + 6 - 2 - \frac{z}{4} = (-3) + (-7) + 6$$

$$-\frac{5z}{8} + 4 - \frac{2z}{8} = -4$$

$$-4 - \frac{7z}{8} + 4 = -4 - 4$$

$$-\frac{7z}{8} = -8$$

$$(8)\left(-\frac{7z}{8}\right) = -8(8)$$

$$-7z = -64$$

$$\frac{-7z}{-7} = \frac{-64}{-7}$$

$$z = \boxed{\frac{64}{7}}$$

$$\textcircled{10} \quad -4 + 7x + 3 - 8x - 1 - 7x = 8 - 2$$

$$-8x - 2 = 6$$

$$2 - 8x - 2 = 6 + 2$$

$$-8x = 8$$

$$\frac{-8x}{-8} = \frac{8}{-8}$$

$$x = \boxed{-1}$$

$$\textcircled{11} \quad 3 - 5 - 3 = 6y + 5y - 7y - 4 + 9$$

$$-5 = 4y + 5$$

$$-5 - 5 = 4y + 5 - 5$$

$$-10 = 4y$$

$$\frac{-10}{4} = \frac{4y}{4}$$

$$\boxed{-\frac{5}{2} = y}$$

$$\textcircled{12} \quad -8 - 5 - \frac{3z}{8} + \frac{z}{6} + 7 - 9 = -3 - 6 - 2$$

$$15 - \frac{9z}{24} + \frac{4z}{24} - 15 = -11 + 15$$

$$-\frac{9z}{24} + \frac{4z}{24} = 4$$

$$-\frac{5z}{24} = 4$$

$$(24)\left(-\frac{5z}{24}\right) = 4(24)$$

$$-5z = 96$$

$$\frac{-5z}{-5} = \frac{96}{-5}$$

$$\boxed{z = -96/5}$$

$$(13) \quad 10 + 9 - 13 + (-1) = 13 + 2z - 4 + 3z - 9z$$

$$5 = -4z + 9$$

$$\textcolor{red}{-9} + 5 = -4z + 9 - 9$$

$$-4 = -4z$$

$$-4 = -4z$$

$$\frac{-4}{-4} = \frac{-4z}{-4}$$

$$\boxed{1 = z}$$

$$(14) \quad 6 - 2x + 8 + (-3) + 10x = -1 - 1 - 1 - 1$$

$$8x + 11 = -4$$

$$\textcolor{red}{-11} + 8x + 11 = -4 - 11$$

$$8x = -15$$

$$\frac{8x}{8} = \frac{-15}{8}$$

$$\boxed{x = -\frac{15}{8}}$$

$$(15) \quad -13 - 16 + 33 = -\frac{y}{2} - \frac{3y}{10} + 7 + \frac{y}{5}$$

$$4 = -\frac{5y}{10} - \frac{3y}{10} + \frac{2y}{10} + 7$$

$$4 = \frac{-6y}{10} + 7$$

$$\textcolor{red}{-7} + 4 = \frac{-6y}{5} + 7 - 7$$

$$-3 = \frac{-6y}{5}$$

$$(5)(-3) = \frac{-6y}{5}(5)$$

$$-15 = -6y$$

$$-15 = -6y$$

$$\frac{-15}{-3} = \frac{-6y}{-3}$$

$$\boxed{5 = y}$$

$$(16) \quad 6x + (-2) - 4x - 7 + x = -8 + (-2) + (-3)$$

$$3x - 9 = -13$$

$$\textcolor{red}{9} + 3x - 9 = -13 + 9$$

$$3x = -4$$

$$\frac{3x}{3} = \frac{-4}{3}$$

$$\boxed{x = -\frac{4}{3}}$$

$$\textcircled{17} \quad -z + 10z + 4z - 1 + 3z = 100 - 101 + (-2)$$

$$16z - 1 = -3$$

$$1 + 16z - 1 = -3 + 1$$

$$16z = -2$$

$$\frac{16z}{16} = \frac{-2}{16}$$

$$z = -\frac{1}{8}$$

$$\textcircled{18} \quad -2 - 10 - 8 - 1 + 4 + (-1) = -\frac{x}{3} + \frac{5x}{9} - 3 - \frac{x}{9}$$

$$-18 = -\frac{3x}{9} + \frac{5x}{9} - \frac{x}{9} - 3$$

$$3 - 18 = \frac{x}{9} - 3 + 3$$

$$-15 = \frac{x}{9}$$

$$(9)(-15) = \frac{x}{9}(9)$$

$$-135 = x$$

$$\textcircled{19} \quad -2 + (-1) - 5x - 2x + 14 + (-3x) = -10 + 15 - 1$$

$$-10x + 11 = 4$$

$$-11 - 10x + 11 = 4 - 11$$

$$-10x = -7$$

$$\frac{-10x}{-10} = \frac{-7}{-10}$$

$$x = \frac{7}{10}$$

$$\textcircled{20} \quad 8 - 4 - 1 + (-2) + (-7) + 20 = x - 2x - 10x + 5$$

$$14 = -11x + 5$$

$$-5 + 14 = -11x + 5 - 5$$

$$9 = -11x$$

$$\frac{9}{-11} = \frac{-11x}{-11}$$

$$-\frac{9}{11} = x$$

$$\star \textcircled{21} \quad -6-8+\frac{z}{2}+(-\frac{7z}{8})+\frac{z}{4}=-8+4-9$$

$$-14+\frac{4z}{8}+(-\frac{7z}{8})+\frac{2z}{8}=-13$$

$$-14-\frac{z}{8}=-13$$

$$14-14-\frac{z}{8}=-13+14$$

$$-\frac{z}{8}=1$$

$$(8)(-\frac{z}{8})=1(8)$$

$$-z=8$$

$$\begin{array}{r} -1z=8 \\ -1 \quad -1 \\ \hline z=-8 \end{array}$$

$$\textcircled{22} \quad 1-9=8-7x-3+3+x-4-7+11$$

$$-8=-6x+8$$

$$-8-8=-6x+8-8$$

$$-16=-6x$$

$$\begin{array}{r} -16=-6x \\ -6 \quad -6 \\ \hline \end{array}$$

$$\boxed{\frac{8}{3}=x}$$

$$\textcircled{23} \quad 18y-14y+6-y+2y+(-8)+2-10y=-2-5$$

$$-5y=-7$$

$$\begin{array}{r} -5y=-7 \\ -5 \quad -5 \\ \hline \end{array}$$

$$\boxed{y=\frac{7}{5}}$$

$$\textcircled{24} \quad -10+8-3-4=-6-\frac{z}{2}-\frac{z}{4}+1-\frac{3z}{4}$$

$$-9=-5-\frac{2z}{4}-\frac{z}{4}-\frac{3z}{4}$$

$$-9=-5-\frac{6z}{4}$$

$$-9=-5-\frac{3z}{2}$$

$$5-9=-5-\frac{3z}{2}+5$$

$$-4=-\frac{3z}{2}$$

$$(2)(-4)=-\frac{3z}{2}(2)$$

$$-8=-3z$$

$$\begin{array}{r} -8=-3z \\ -3 \quad -3 \\ \hline \end{array}$$

$$\boxed{\frac{8}{3}=z}$$

$$(25) \quad 11x + 7x - 3 + 8 - 2x = -7 + 10 - 1$$

$$16x + 5 = 2$$

$$-5 + 16x + 5 = 2 - 5$$

$$16x = -3$$

$$\frac{16x}{16} = \frac{-3}{16}$$

$$x = -\frac{3}{16}$$

$$(26) \quad -8 + (-8) + 4 + 3 - 2 - 2 = 7x - 12x + 3x + 10 - 1$$

$$-13 = -2x + 9$$

$$-9 - 13 = -2x + 9 - 9$$

$$-22 = -2x$$

$$\frac{-22}{-2} = \frac{-2x}{-2}$$

$$11 = x$$

$$(27) \quad 8 - x + 2 - 6x - 7 + 9x = -5 + 15 + (-10)$$

$$2x + 3 = 0$$

$$-3 + 2x + 3 = 0 - 3$$

$$2x = -3$$

$$\frac{2x}{2} = \frac{-3}{2}$$

$$x = -\frac{3}{2}$$

$$(28) \quad 15 - 5 + 6 - 3 - 3 - 1 + (-9) = 10 - 7x + (-2) + x$$

$$0 = -6x + 8$$

$$-8 + 0 = -6x + 8 - 8$$

$$-8 = -6x$$

$$\frac{-8}{-6} = \frac{-6x}{-6}$$

$$\frac{4}{3} = x$$